Optimal Stopping Formulation Coupling Entry Tear Diameter False Lumen Expansion and Serial Enlargement for Preemptive TEVAR After Type B Dissection

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Abstract

Uncomplicated type B aortic dissection (TBAD) may remain stable or undergo progressive false-lumen and aortic enlargement, while the timing of preemptive thoracic endovascular aortic repair (TEVAR) remains uncertain. We developed a literature-informed mathematical proof-of-concept that formulates continued surveillance versus TEVAR as a finite horizon optimal stopping problem. The model combines two observable morphological states false-lumen diameter $F(t)$ and total aortic diameter $D(t)$ with three model-defined latent states representing growth momentum G(t), morphologic burden $B(t)$, and stability reserve $R(t)$. Entry tear diameter, false lumen diameter, aortic diameter, serial enlargement, and disease phase definitions were used as literature derived normalization or contextual anchors rather than fitted treatment thresholds. Structural and decision coefficients were provisional and were not estimated from patient-level outcomes. Continuous state dynamics were coupled to a four-regime continuous time Markov representation and competing theoretical waiting and procedural intensities, with a Bellman recursion defining continuation and stopping regions. Computational analyses comprised free boundary continuation, post-TEVAR equilibrium and Jacobian stability analysis, Latin hypercube parameter sets with partial rank correlation coefficients, stochastic input perturbations, numerical-solver verification, and an internal synthetic model-recovery experiment. Under the prespecified reference simulation (12-mm entry tear, 22-mm false lumen, and 41-mm total aortic diameter), the model produced a first stopping boundary at day 33. Across the explored entry-tear range of 6-16 mm and false lumen range of 16-30 mm, model-derived stopping boundaries ranged from 25 to 47 days. Stochastic perturbations produced a median boundary of 33 days (2.5^{th}-97.5^{th} percentiles, 27-41 days), while the post-TEVAR equilibrium yielded five negative real eigenvalues (-0.140 to -0.010). The strongest PRCCs with the simulated stopping day were observed for waiting intensity (-0.932), acute procedural penalty (0.924), and baseline procedure intensity (0.917). These quantities describe the behavior of the specified mathematical system and are not estimates of patient-level treatment effects, clinical risks, or optimal clinical timing. The model provides a reproducible structure for investigating how evolving CTA morphology could alter a surveillance versus intervention boundary and for specifying variables required for subsequent clinical calibration. Patient level longitudinal calibration, external validation, and prospective evaluation are required before any stopping boundary can be interpreted as clinical decision support.

Keywords*: Entry Tear, False Lumen Expansion, Optimal Stopping, Preemptive TEVAR, Serial Aortic Enlargement, Type B Aortic Dissection*

1. Introduction

Uncomplicated type B aortic dissection is not a biologically uniform surveillance state. Contemporary guidelines recognize high-risk imaging features such as a large primary entry tear, an enlarged false lumen, increased total aortic diameter, and enlargement on serial imaging, while randomized and observational evidence suggests that TEVAR can promote favorable remodeling in selected patients. The remaining clinical problem is not simply whether these features are associated with later events. It is how an evolving morphology should change the value of continued surveillance relative to the value and procedural cost of preemptive TEVAR. The current condition therefore treats the problem as a sequential decision under uncertainty: the dissected aorta evolves over time, procedural risk also changes with phase, and an intervention boundary is reached when the modeled cost of waiting no longer remains favorable.

The financial engineering contribution is methodological rather than metaphorical. Optimal stopping, free-boundary analysis, dynamic reserve, regime switching, competing hazards, stochastic perturbation, and uncertainty propagation are mature quantitative tools for decisions in systems whose state and action costs evolve over time. Here, they are transferred to aortic morphology without treating a patient as a financial asset. The model is deliberately falsifiable: future serial CTA and outcome data may calibrate, modify, or reject any component.

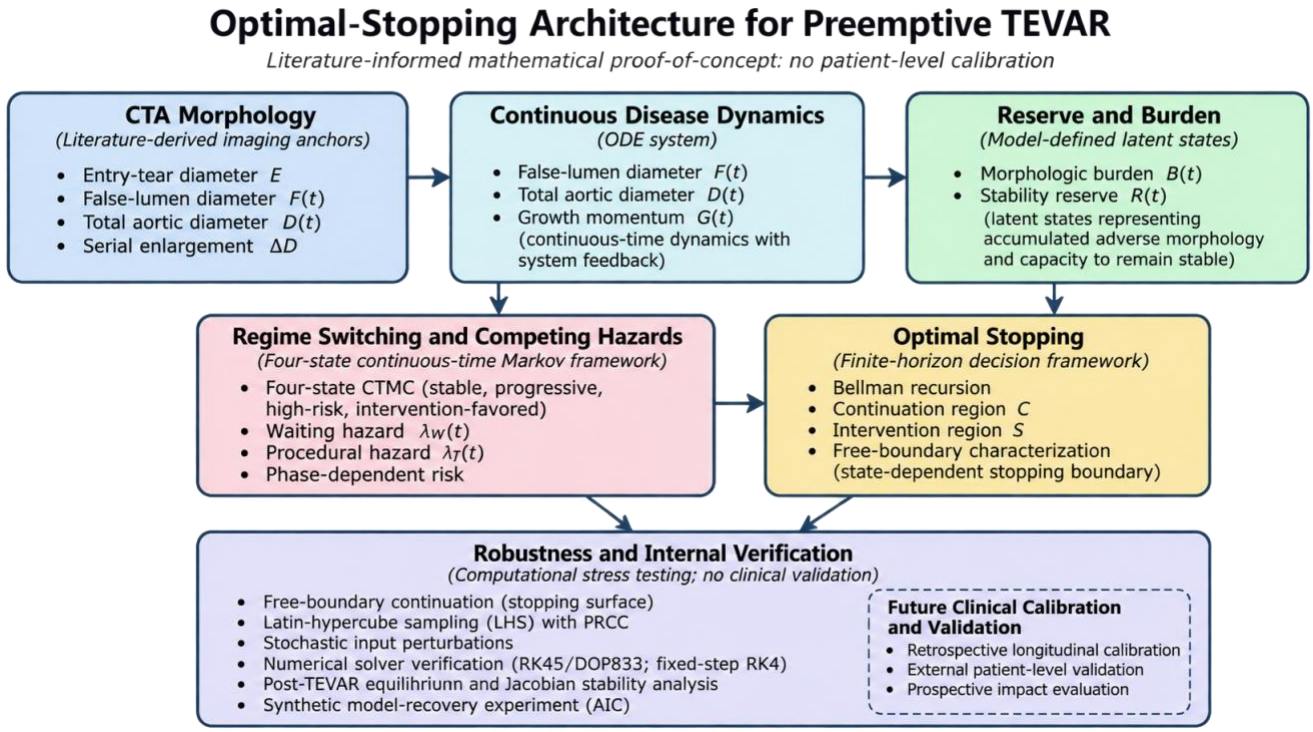


Figure 1. *Hybrid architecture linking CTA morphology, continuous disease dynamics, competing hazards, and the optimal-stopping layer.*

2. Methods

2.1.Study Design and Evidence Status

This study developed a literature-informed mathematical proof-of-concept for representing the timing of preemptive thoracic endovascular aortic repair (TEVAR) in uncomplicated type B aortic dissection (TBAD) as a sequential optimal stopping problem. The analysis was designed as structural model development and computational stress testing rather than as clinical prediction model development or estimation of a treatment effect. No individual level longitudinal clinical outcome dataset was used for model fitting, parameter estimation, or validation. Consequently, the numerical stopping boundaries based on this study should be interpreted as outputs of the specified mathematical system rather than estimates of clinically optimal TEVAR timing.

The observable morphological inputs were selected a priori because they can be obtained from serial computed tomography angiography (CTA) and have been repeatedly considered in guideline-based and longitudinal assessments of TBAD progression. These included primary entry tear diameter, false-lumen diameter, total aortic diameter, and change in aortic dimensions on serial imaging. Contemporary ACC/AHA and ESC guidance identifies morphological characteristics and progressive aortic enlargement as relevant features for risk assessment and surveillance in aortic dissection (Isselbacher et al., 2022 and Mazzolai et al., 2024). Previous longitudinal studies have additionally reported associations between entry tear characteristics, false lumen morphology, and subsequent aortic enlargement or adverse remodeling (Evangelista et al., 2012 and Trimarchi et al., 2014). Evidence from randomized studies of endovascular treatment, including INSTEAD-XL and ADSORB, was used to provide clinical context for a surveillance versus intervention formulation, but not to derive patient specific treatment effects or model coefficients (Nienaber et al., 2013 and Brunkwall et al., 2014). Published analyses of acute and subacute intervention timing were likewise used only to contextualize disease phase definitions and timing assumptions (Yang et al., 2022 and Naito and Takagi, 2026).

Accordingly, the model distinguished three levels of evidence. First, literature derived quantities, including morphological and disease phase sources, were used for normalization and clinical context and were not treated as fitted treatment thresholds. Second, observable state variables represented quantities that could in principle be measured longitudinally from CTA. Third, latent state dynamics, transition parameters, theoretical waiting and procedural intensities, and Bellman cost coefficients were model defined quantities introduced to construct the proof of concept decision system. These structural and decision coefficients were not inferred from published odds ratios, hazard ratios, or individual patient records. They were assigned prespecified reference values and uncertainty ranges for computational exploration, with their influence subsequently examined through local perturbation, Latin hypercube sampling, partial rank correlation analysis, and stochastic stress testing.

A publicly available expert annotated TBAD CTA collection comprising 40 cases was identified as a potential resource for future evaluation of image-derived morphological measurements. It was not used for calibration or validation of the present stopping model because the dataset does not provide the longitudinal combination of serial morphology, treatment timing, and patient level outcomes required to estimate the proposed dynamic and decision components. Clinical calibration of the model therefore

remains a separate future stage requiring longitudinal patient-level data with serial CTA measurements, treatment dates, and relevant outcomes.

2.2.Literature-Informed Model Anchors

Literature-informed morphological and temporal anchors were used to place the mathematical state variables on clinically interpretable scales. These values were selected from guideline and longitudinal TBAD literature describing imaging features associated with higher risk morphology, aortic enlargement, or disease-phase classification (Trimarchi et al., 2014 and Isselbacher et al., 2022). They served as normalization or contextual reference values within the mathematical model and were not estimated from the present simulations. Importantly, these anchors were not treated as validated patient specific treatment thresholds, causal effect estimates, or stand-alone indications for TEVAR.

Table 1. Literature-Informed Anchors for Model Normalization and Temporal Context

Domain	Literature-informed anchor	Role in mathematical Model	Evidence basis
Primary entry tear diameter	>10 mm	Morphological reference and normalization anchor	Evangelista et al. (2012), Isselbacher et al. (2022)
False lumen diameter	approximately 20-22 mm	Reference range for false lumen morphology and normalization	Trimarchi et al. (2014), Isselbacher et al. (2022)
Maximum total aortic diameter	>40 mm	Reference anchor for increased aortic size	Isselbacher et al. (2022), Mazzolai et al. (2024)
Serial aortic enlargement	>5 mm between serial assessments	Longitudinal change reference	Isselbacher et al. (2022), Mazzolai et al. (2024)
Disease phase	Acute <14 days, subacute 14-90 days	Context for phase dependent model terms	Yang et al. (2022), Mazzolai et al. (2024), Naito & Takagi (2026)

2.3. State Variables

The continuous state vector was defined as

$$X(t) = [F(t), D(t), G(t), B(t), R(t)]$$

The five states were introduced to distinguish directly interpretable morphological quantities from latent mathematical quantities required to represent temporal dependence within the dynamical system. False lumen diameter $F(t)$ and total aortic diameter D(t) correspond to morphological quantities that can, in principle, be measured repeatedly from CTA. Their inclusion was motivated by longitudinal TBAD literature describing false-lumen morphology and progressive aortic enlargement as relevant features of disease evolution (Trimarchi et al., 2014; Isselbacher et al., 2022 and Mazzolai et al., 2024).

In contrast, growth momentum $G(t)$, morphologic burden $B(t)$, and stability reserve $R(t)$ were introduced as dimensionless, model defined latent states. They were not directly measured in patients, extracted from the cited CTA literature, or fitted to observed clinical outcomes. $G(t)$ was used to retain information about the recent tendency of morphological progression within the dynamical system: $B(t)$ summarized accumulated adverse morphological pressure made by the model: and $R(t)$ represented a normalized mathematical reserve variable that decreases as the modelled system becomes less favourable

to continued surveillance. Their values therefore have meaning only within the specified mathematical model and should not be interpreted as validated biomarkers, physiological measurements, clinical scores, or patient-specific risk probabilities.

All dimensionless latent states were represented on a normalized scale from 0 to 1 for computational interpretation and bounded model behaviour. This normalization does not imply that equal numerical changes in $G(t), B(t)$, or $R(t)$ correspond to equal clinical changes. Their dynamics, initial conditions, and coupling coefficients were specified provisionally and were subsequently examined through sensitivity and uncertainty analyses rather than treated as empirically estimated quantities.

Table 2. Observable and Model Defined State Variables in the Dynamical Model

Symbol	State	Scale	Evidence	Interpretation the model
$F(t)$	False lumen diameter	mm	Potentially observable from serial CTA	Time varying false lumen dimension represented within the continuous dynamical system
$D(t)$	Total aortic diameter	mm	Potentially observable from serial CTA	Time varying total aortic dimension used to represent serial aortic enlargement
$G(t)$	Growth momentum	0-1	Model defined latent state	Dimensionless memory state representing the modelled tendency for ongoing morphological progression
$B(t)$	Morphologic burden	0-1	Model defined latent state	Dimensionless summary of accumulated adverse morphological pressure within the model
$R(t)$	Stability reserve	0-1	Model defined latent state	Dimensionless mathematical state representing the modelled capacity of the system to remain favourable to continued surveillance

2.4. Mathematical Formulation

The mathematical model was constructed as a layered dynamical decision system rather than as a single fitted clinical prediction equation. The formulation comprised: (i) literature-informed transformations of observable morphology, (ii) continuous pre-intervention state dynamics, (iii) a post-TEVAR dynamical subsystem; (iv) a state dependent continuous time Markov representation, (v) theoretical waiting and procedural intensities, and (vi) a finite horizon Bellman recursion defining continuation and intervention regions. This separation was intentional because the components represent different mathematical roles and levels of empirical support. None of the structural coefficients, theoretical intensities, or decision costs described below were estimated from patient level outcomes.

2.4.1. Morphological Normalization

Entry tear diameter E, false lumen diameter $F(t)$, and total aortic diameter $D(t)$ were transformed to dimensionless quantities before entering the latent state and intensity functions:

$$e(E) = \left\{1 + \exp\left[-\frac{E-10}{2}\right]\right\}^{-1},$$

$$f(F) = \text{clip}\left(\frac{F-15}{20}, 0, 1.5\right),$$

$$d(D) = \text{clip}\left(\frac{D-35}{25}, 0, 1.5\right),$$

These transformations were introduced for numerical scaling and to provide smooth or bounded representations of increasing morphological burden. They should not be interpreted as empirically estimated dose response functions or clinical risk probabilities. Their centering and scaling were informed by the morphological reference ranges described in Section 2.2, including literature concerning entry tear characteristics, false lumen morphology, and aortic enlargement (Trimarchi et al., 2014; Isselbacher et al., 2022 and Mazzolai et al., 2024). The exact functional forms and scaling constants are model design choices and therefore remain provisional.

2.4.2. Pre-Intervention Continuous State Dynamics

Before intervention, the state vector

$$X(t) = [F(t), D(t), G(t), B(t), R(t)]$$

was evolved using the following coupled ordinary differential equations:

$$\frac{dF}{dt} = k_{F0} + k_{FE}e(E)\left(1 - \frac{F}{50}\right) + k_{FG}G - k_{Fr}\frac{\max(F-18,0)}{30},$$

$$\frac{dD}{dt} = k_{D0} + k_{DF}f(F) + k_{DG}G - k_{Dr}\frac{\max(D-38,0)}{30},$$

$$\frac{dG}{dt} = [k_{GE}e(E) + k_{GF}f(F) + k_{GD}d(D)](1-G) - k_{Gr}\,G,$$

$$\frac{dB}{dt} = [k_{BE}e(E) + k_{BF}f(F) + k_{BG}G](1-B) - k_{Br}RB,$$

$$\frac{dR}{dt} = k_R(1-R) - k_{RB}BR - k_{RG}GR$$

The equations were constructed to encode a qualitative feedback architecture rather than an empirically identified biological mechanism. Larger modeled entry tear and false-lumen states increase progression-related terms, while the saturating factors constrain the normalized latent states. $G(t), B(t)$, and $R(t)$ retain the definitions given in Section 2.3 and are model-defined latent variables rather than directly observed physiological quantities.

The coefficients $k_{\cdot}$ are therefore provisional structural parameters. Their reference values were selected to produce bounded and directionally interpretable trajectories within the explored state domain; they were not estimated from clinical event rates, regression coefficients, hazard ratios, or patient trajectories. Their influence on the resulting stopping boundary was evaluated subsequently through local and global sensitivity analyses.

2.4.3. Post-TEVAR Dynamical Subsystem

To represent the mathematical consequences of switching from continued surveillance to intervention, a separate post-TEVAR subsystem was specified:

$$\left.\frac{dF}{dt}\right|_{\text{TEVAR}} = -k_{TF}\left(F - F_{\text{target}}\right),$$

$$\left.\frac{dD}{dt}\right|_{\text{TEVAR}} = -k_{TD}\left(D - D_{\text{target}}\right),$$

$$\left.\frac{dG}{dt}\right|_{\text{TEVAR}} = -k_{TG}G,$$

$$\left.\frac{dB}{dt}\right|_{\text{TEVAR}} = -k_{TB}B\ \max(R, 0.15),$$

$$\left.\frac{dR}{dt}\right|_{\text{TEVAR}} = k_{TR}(1 - R) - 0.015\ BR.$$

This subsystem was designed to represent a hypothetical transition toward a lower burden post intervention state and to permit equilibrium and local stability analysis. It was motivated qualitatively by evidence that TEVAR can promote favorable aortic remodeling in selected TBAD populations (Nienaber et al., 2013 and Brunkwall et al., 2014), but the equations, target values, and rate coefficients were not fitted to remodeling trajectories from those trials. Accordingly, the post-TEVAR dynamics should be interpreted as a structural component of the proof-of-concept rather than a quantitative model of an individual patient's response to TEVAR.

2.4.4. Regime Switching Representation

A four-state continuous time Markov chain was used to represent transitions among model defined regimes corresponding to stable, progressive, high risk, and intervention favored states. If

$$p(t) = [p_1(t), p_2(t), p_3(t), p_4(t)]$$

denotes the regime probability vector, its evolution was represented by

$$p'(t) = p(t)Q[X(t)],$$

where $Q[X(t)]$ is a state dependent generator matrix. The Markov layer was included to represent discrete changes in modeled disease regime alongside the continuous morphological dynamics. These regime probabilities were not estimated from observed patient transitions and therefore should not be interpreted as empirical probabilities of clinical progression. The transition structure is a theoretical component to be calibrated when longitudinal state-transition data become available.

2.4.5. Theoretical Waiting and Procedural Intensities

Two state dependent intensities were introduced to summarize the modeled pressure associated with continued surveillance and intervention:

$$\lambda_W(t) = h_{W0}\ \exp[b_E e(E) + b_F f(F) + b_D d(D) + b_G G + b_R(1 - R)],$$

and

$$\phi(t) = 1 + a\ \exp\left(-\frac{t}{\tau_a}\right) + c\left\{1 + \exp\left[-\frac{t - t_c}{s_c}\right]\right\}^{-1},$$

$$\lambda_{T(t)} = h_{T0}\ \phi(t)\exp[c_F f(F) + c_D d(D) + c_R(1 - R)].$$

Here, $\lambda_W(t)$ and $\lambda_T(t)$ are theoretical model intensities, not empirically estimated clinical hazard functions. The phase modifier $\phi(t)$ allows the modeled procedural contribution to vary with time after dissection, reflecting the general clinical motivation that disease phase is relevant to intervention timing (Yang et al., 2022; Mazzolai et al., 2024 and Naito and Takagi, 2026). However, neither $\phi(t)$ nor its

coefficients should be interpreted as an estimated temporal risk curve for TEVAR. This distinction is essential. The intensity functions provide internally consistent inputs to the decision layer but do not represent calibrated probabilities of mortality, rupture, complications, or treatment benefit.

2.4.6. Finite Horizon Optimal Stopping

The surveillance versus intervention decision was formulated as a finite horizon stopping problem using dynamic programming principles, in which immediate intervention is compared with the value of continuation at each decision state (Shiryaev, 2008 and Bäuerle & Rieder, 2011). The surveillance versus intervention decision was formulated using a finite horizon Bellman recursion:

$$V_k(x) = \min\{ C_{\text{TEVAR}}(x,k), C_{\text{wait}}(x,k) + \gamma V_{k+1}[T(x)]\},$$

where $C_{\text{TEVAR}}(x,k)$ denotes the model-defined cost of intervention at state x and decision time k, $C_{\text{wait}}(x,k)$ denotes the corresponding cost of continued surveillance, $T(x)$ represents state propagation to the subsequent decision step, and γ is the continuation weighting factor. The first model derived stopping time was defined as:

$$\tau^* = \inf\{ t \geq 0 : C_{\text{TEVAR}}(X_t, t) \leq C_{\text{wait}}(X_t, t) + \gamma V_{k+1}\}.$$

The corresponding continuation and intervention regions were

$$C = \{x : \text{continued surveillance is model optimal}\}$$

$$S = \{x : \text{intervention is model optimal}\}$$

with ∂S denoting the free boundary separating the two decision regions. The Bellman costs are dimensionless decision constructs used to compare the two-actions within the mathematical model. They are not monetary costs, quality adjusted life years, observed complication rates, or empirically estimated utilities. Consequently, τ^* represents the first stopping boundary made by the specified model and parameterization, not a recommended calendar time for TEVAR.

2.4.7. Interpretation and Evidence Boundary

The formulation deliberately separates quantities supported by existing observations from quantities introduced to construct the mathematical decision system. Entry tear diameter, false lumen diameter, total aortic diameter, serial enlargement, and disease phase definitions are linked to existing TBAD literature (Isselbacher et al., 2022; Mazzolai et al., 2024 and Naito and Takagi, 2026). In contrast, the latent states $G(t), B(t)$, and $R(t)$, structural coefficients, Markov transition rates, theoretical intensities $\lambda_W(t)$ and $\lambda_T(t)$, and Bellman cost parameters are provisional model constructs. Accordingly, the present formulation tests whether these components can make a coherent, reproducible, state-dependent optimal-stopping structure under explicitly stated assumptions. It does not establish that the assumed equations represent the true biological evolution of TBAD or that the resulting stopping boundary is clinically optimal. Those claims would require longitudinal patient level calibration, identifiability analysis, observed outcome validation, and external evaluation.

2.5 Theoretical Analysis

Local existence and uniqueness were considered over the bounded simulation domain. The pre- and post-TEVAR vector fields are piecewise smooth and locally Lipschitz, including at the $\max(\cdot)$ and

clipping boundaries. The states remained within the specified computational domain across the simulations, no claim of global invariance is made. The post-TEVAR equilibrium was solved numerically, and local asymptotic stability was assessed from the eigenvalues of the Jacobian evaluated at that equilibrium. Entry tear and false lumen continuation was treated as free boundary analysis rather than classical bifurcation analysis because the object of interest was the Bellman decision boundary, not an equilibrium transition. Structural identifiability was not assessed because no patient-level observation model was fitted.

2.6 Computational and Decision Analysis

Reference trajectories were simulated over 120 days using adaptive Runge-Kutta integration, with fixed-step RK4 used for computationally intensive grid and sensitivity analyses. The reference state comprised a 12-mm entry tear, 22-mm false lumen, 41-mm total aortic diameter, $G(0) = 0.20, B(0) = 0.15$, and $R(0) = 0.78$. The morphological values defined a prespecified reference configuration, whereas the latent state values were model defined initial conditions. Numerical integration and solver cross-checking followed standard principles of numerical solution of ordinary differential equations and scientific computing (Press et al., 2007). The Bellman recursion compared continued surveillance with immediate TEVAR at daily decision steps, and the first intervention favoured day was recorded as the model-derived stopping boundary. Free boundary continuation varied entry-tear diameter from 6-16 mm and initial false lumen diameter from 16-30 mm, including a 609-point two dimensional surface. Eight prespecified model scenarios additionally examined directional changes across morphology and latent-state configurations. All stopping times were simulation outputs rather than clinical treatment recommendations.

2.7 Sensitivity, Uncertainty, and Computational Verification

Global parameter uncertainty was explored using Latin-hypercube sampling, a stratified sampling strategy for efficient exploration of multidimensional input spaces (McKay et al., 1979). Global parameter uncertainty was evaluated using 500 Latin-hypercube parameter sets across prespecified structural and decision-parameter ranges. Parameter influence on the stopping day was summarized using partial rank correlation coefficients (PRCCs) after rank residualization. Local robustness was assessed using $\pm 1\%$ single-parameter perturbations. A separate 500-run stochastic experiment perturbed entry tear diameter, false lumen diameter, total aortic diameter, growth momentum, and stability reserve around the reference state. These simulations were computational stress tests and were not intended to represent an empirical patient distribution. Numerical robustness was assessed by comparing RK45 and DOP853 across output intervals of 0.10-1.00 day. Finally, an internal model recovery experiment fitted five nested logistic specifications to 800 model-based synthetic states and compared them using AIC. Because the labels were made from the same mathematical architecture, this experiment evaluated internal code/model recoverability only and was not used as clinical model selection or predictive validation.

3. Results

3.1. Reference Morphologic Trajectory and Stopping Point

Under the prespecified reference configuration, the Bellman decision rule produced a first model-derived stopping boundary at day 33. By day 30, false-lumen diameter increased from 22.0 to 23.57 mm and total aortic diameter from 41.0 to 41.84 mm, while stability reserve decreased to 0.359. These values are outputs of the reference simulation and do not represent observed serial CTA measurements.

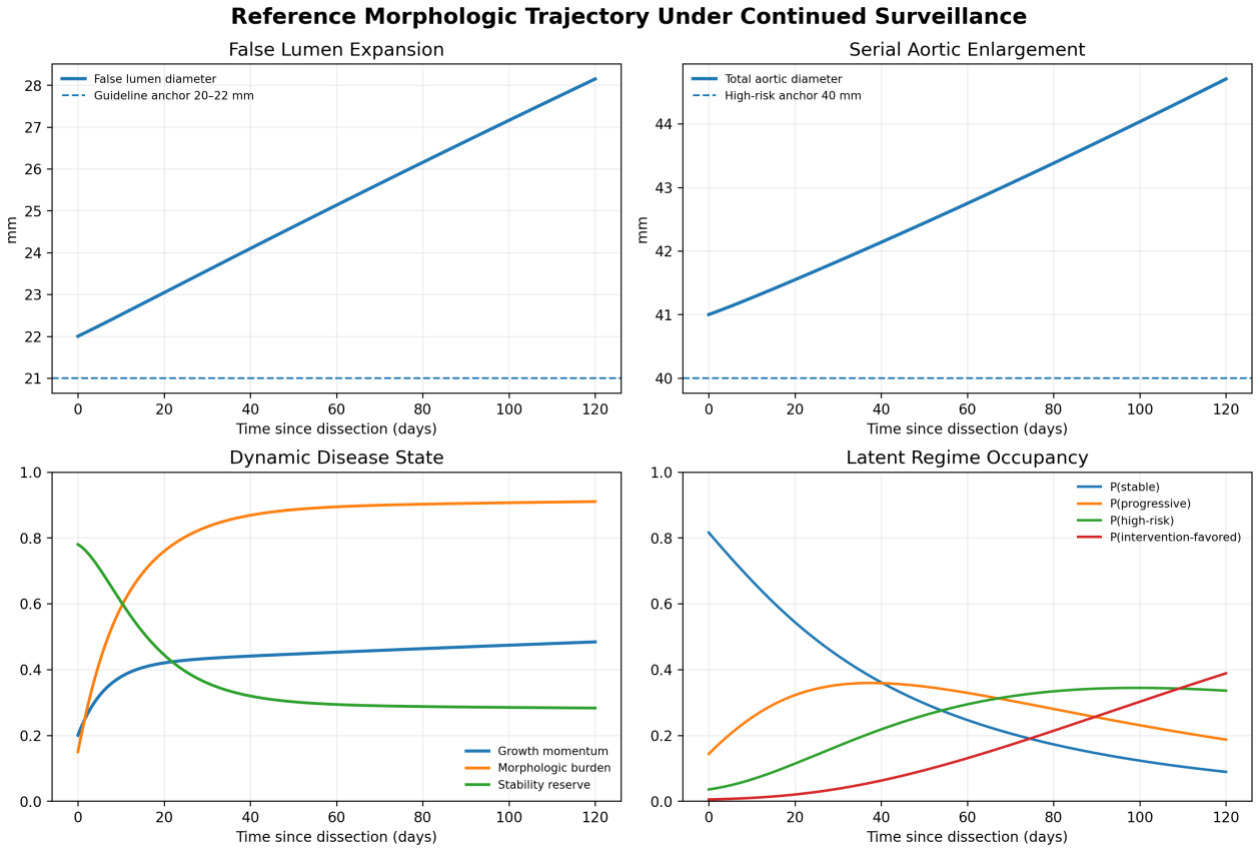


Figure 2. *Simulated reference trajectory under continued surveillance. Dashed lines indicate literature-informed morphological anchors rather than fitted outcome thresholds.*

The theoretical waiting and procedural intensities evolved differently over the simulation horizon, and the Bellman comparison first favored the intervention branch at day 33. This crossing defines the stopping boundary for the reference parameterization and is not a recommended clinical treatment date.

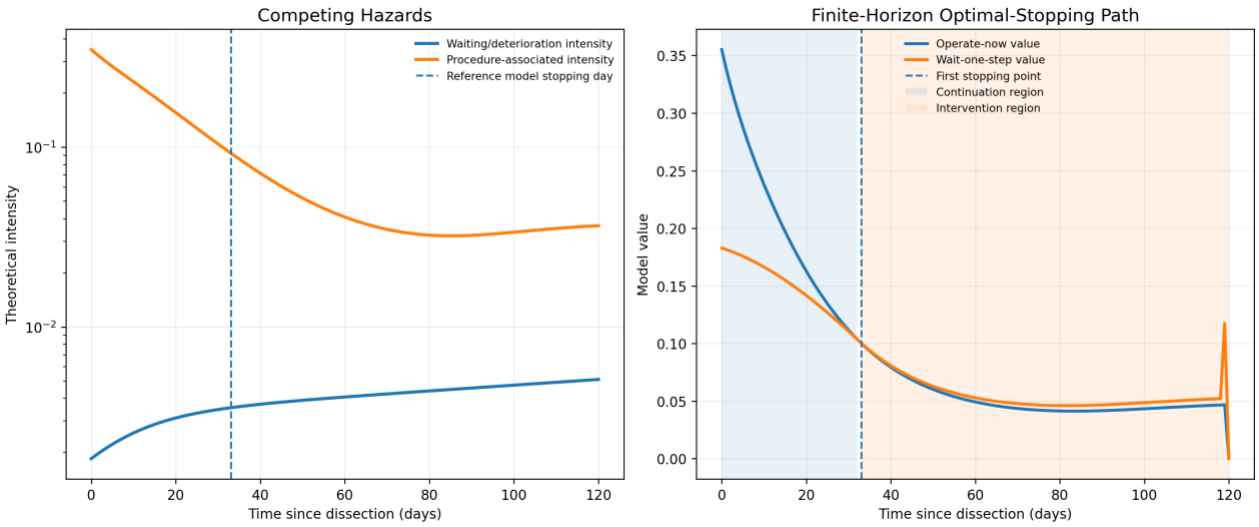


Figure 3. *Theoretical model intensities and finite-horizon optimal-stopping path. The dashed vertical line marks the first model derived stopping boundary.*

3.2. Post-TEVAR Equilibrium and Local Stability

The post-TEVAR remodeling equilibrium was

$$F^* = 17.0\ \text{mm}, \qquad D^* = 39.0\ \text{mm}, \qquad G^* = 0.000, \qquad B^* \approx 0, \qquad R^* = 1.000.$$

The five Jacobian eigenvalues were $-0.140, -0.075, -0.045, -0.022,$ and -0.010, with zero imaginary components. Because all real parts were negative, the equilibrium was locally asymptotically stable under the specified post-TEVAR subsystem. This is a property of the mathematical system and does not establish clinical post-TEVAR stability.

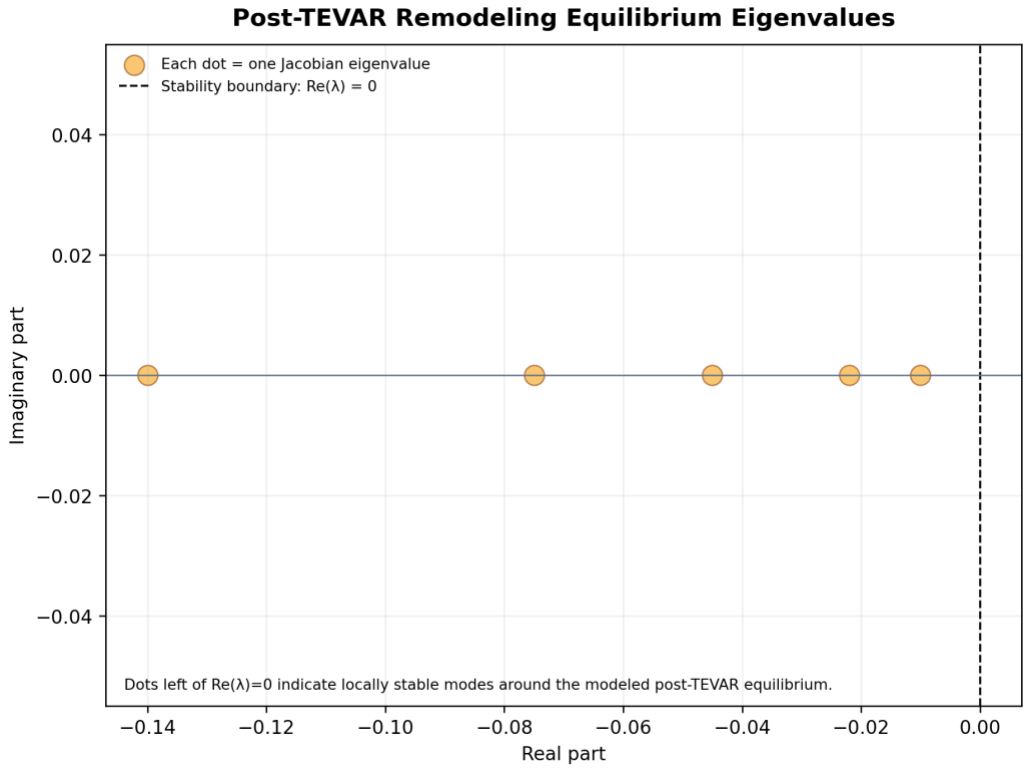


Figure 4. *Jacobian eigenvalues at the modeled post-TEVAR equilibrium. All eigenvalues have negative real parts under the reference parameterization.*

3.3. Free Boundary Continuation and Morphology Dependent Timing

Across the 609-point grid spanning entry-tear diameters of 6-16 mm and initial false lumen diameters of 16-30 mm, model derived stopping days ranged from 25 to 47 days, with a median of 35 days. Within the explored grid, larger entry tear and false lumen dimensions generally shifted the stopping boundary earlier, whereas smaller dimensions extended the continuation region.

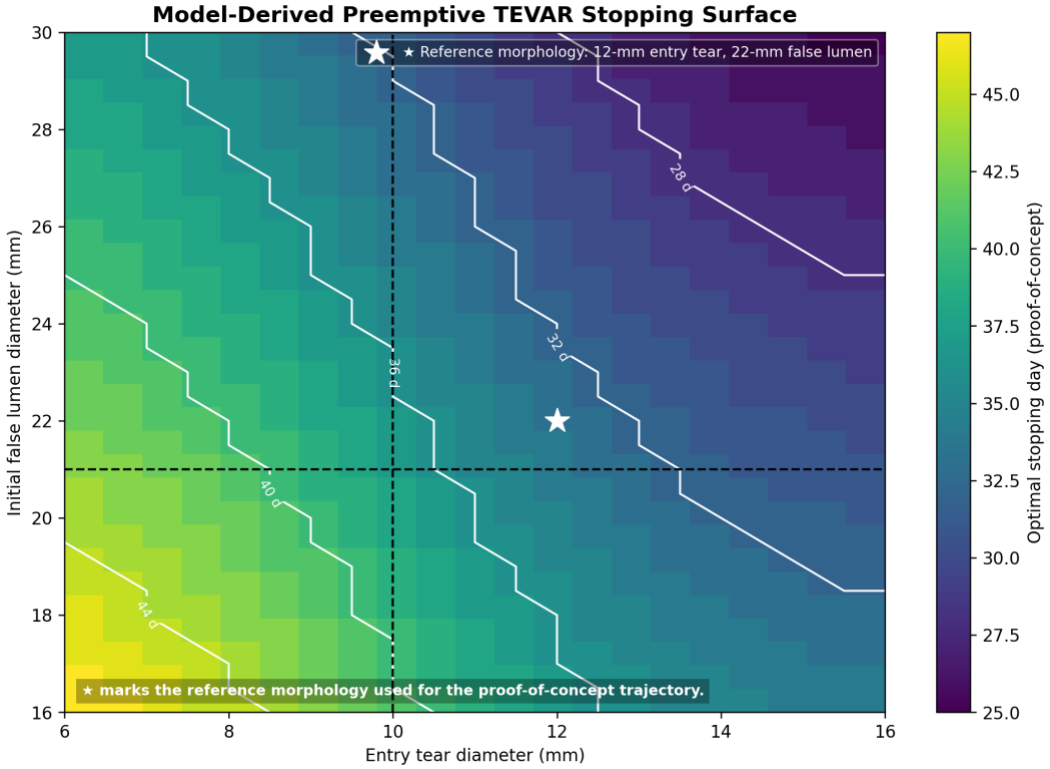


Figure 5. *Model-derived stopping surface across entry tear diameter and initial false lumen diameter. The star indicates the reference morphology (12-mm entry tear, 22-mm false lumen).*

Free boundary continuation across four prespecified false lumen levels showed the same directional pattern, with stopping boundaries generally occurring earlier as entry tear diameter increased. The shaded 14-90-day interval provides literature-informed temporal context only and is not a validation interval.

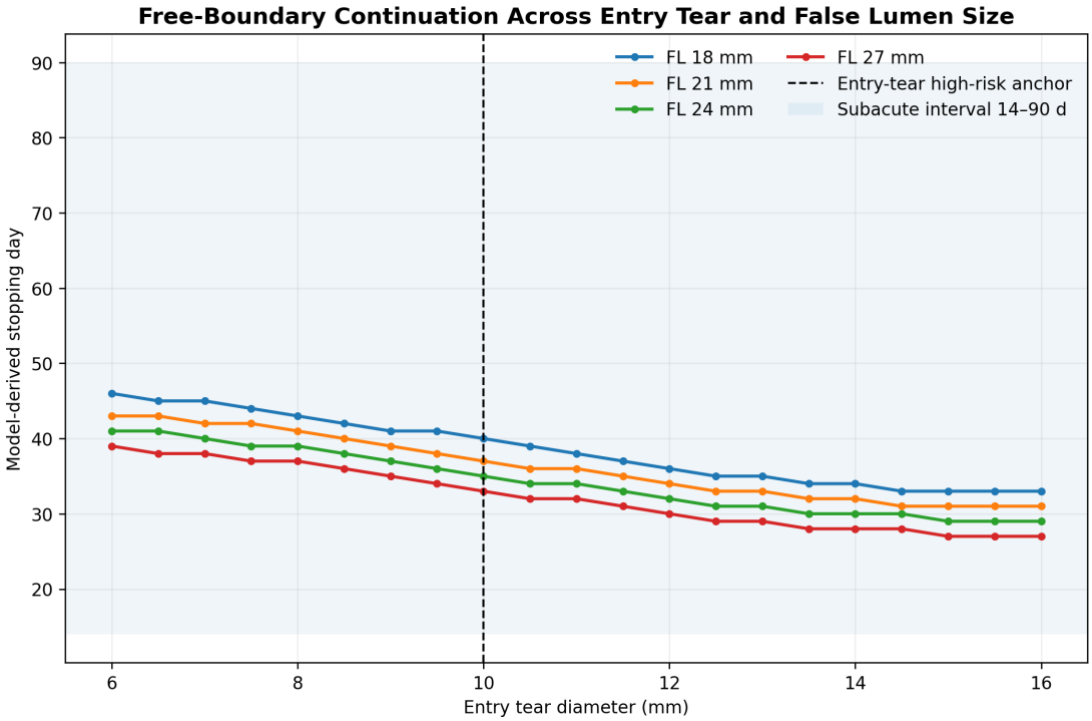


Figure 6. *Free boundary continuation across entry tear diameter at four initial false lumen levels. The shaded 14-90-day interval is a literature-informed phase reference.*

3.4. Prespecified Scenario Stress Testing

The eight prespecified model scenarios produced directionally ordered changes in the stopping boundary. Combined high-risk morphology produced the earliest boundary at 26 days, followed by large entry tear and expanded false lumen at 30 days, serial enlargement burden at 31 days, and the reference configuration at 33 days. Borderline morphology produced a boundary at 38 days, while the smaller tear or false lumen scenario produced the latest boundary at 43 days. The low stability reserve scenario remained at 33 days on the daily decision grid. These scenarios are deterministic model stress tests rather than observed patient groups.

Table 3. Prespecified Scenario Outputs

Scenario	Entry tear (mm)	FL baseline (mm)	Aorta baseline (mm)	Model-derived stopping day
Reference high risk morphology	12	22	41	33
Smaller tear and false lumen	8	18	39	43
Large entry tear	15	22	41	30
Expanded false lumen	12	27	43	30
Serial enlargement burden	12	24	45	31
Low stability reserve	12	22	41	33
Combined high risk morphology	15	28	45	26
Borderline morphology	10	20	40	38

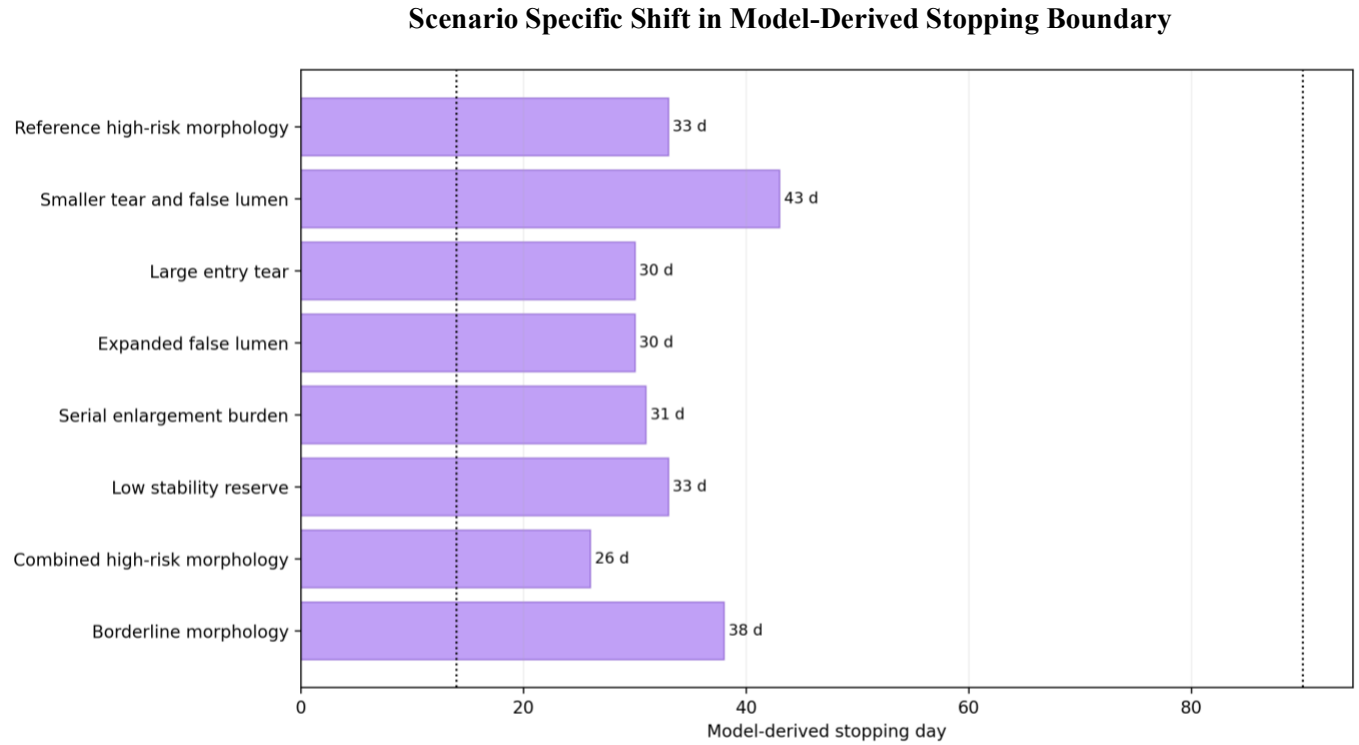


Figure 7. *Model-derived stopping boundaries across eight prespecified scenario stress tests.*

3.5. Parameter Sensitivity and Stochastic Input Perturbation

Across 500 Latin-hypercube parameter sets, the model-derived stopping day had a median of 31 days and a 2.5^{th} - 97.5^{th} percentile range of 11-50.5 days. The largest absolute PRCCs were observed for baseline waiting intensity -0.932, acute procedural penalty 0.924, baseline procedure intensity 0.917, tear driven growth momentum -0.265, false lumen driven momentum -0.159, and reserve recovery 0.143. These coefficients quantify sensitivity within the prespecified parameter ranges and are not clinical effect estimates.

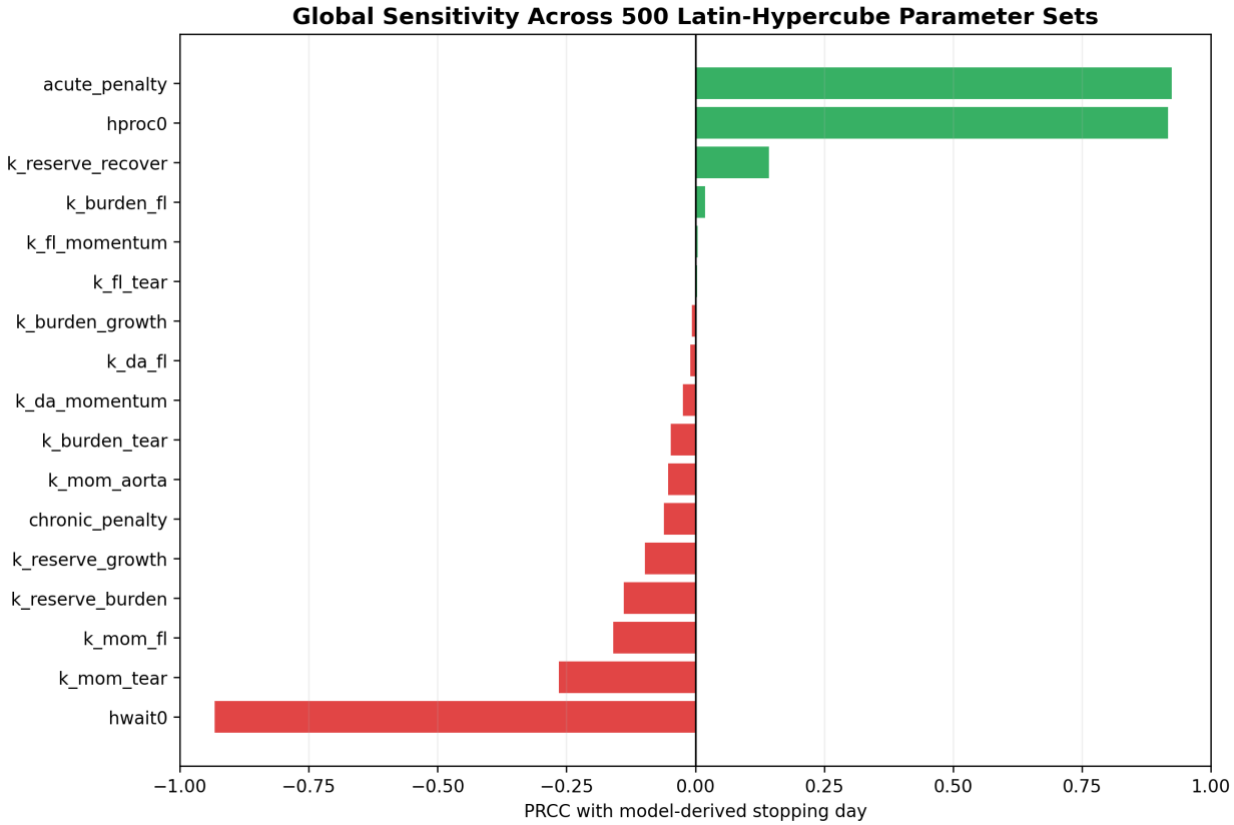


Figure 8. *PRCC sensitivity of the model derived stopping day across 500 Latin-hypercube parameter sets.*

Across 500 stochastic input perturbations around the reference configuration, the median stopping day was 33 days, with a 2.5^{th} - 97.5^{th} percentile range of 27-41 days. These distributions represent computational input perturbations rather than empirical patient variability. Individual $\pm 1\%$ parameter perturbations did not change the discretized 33-day reference boundary.

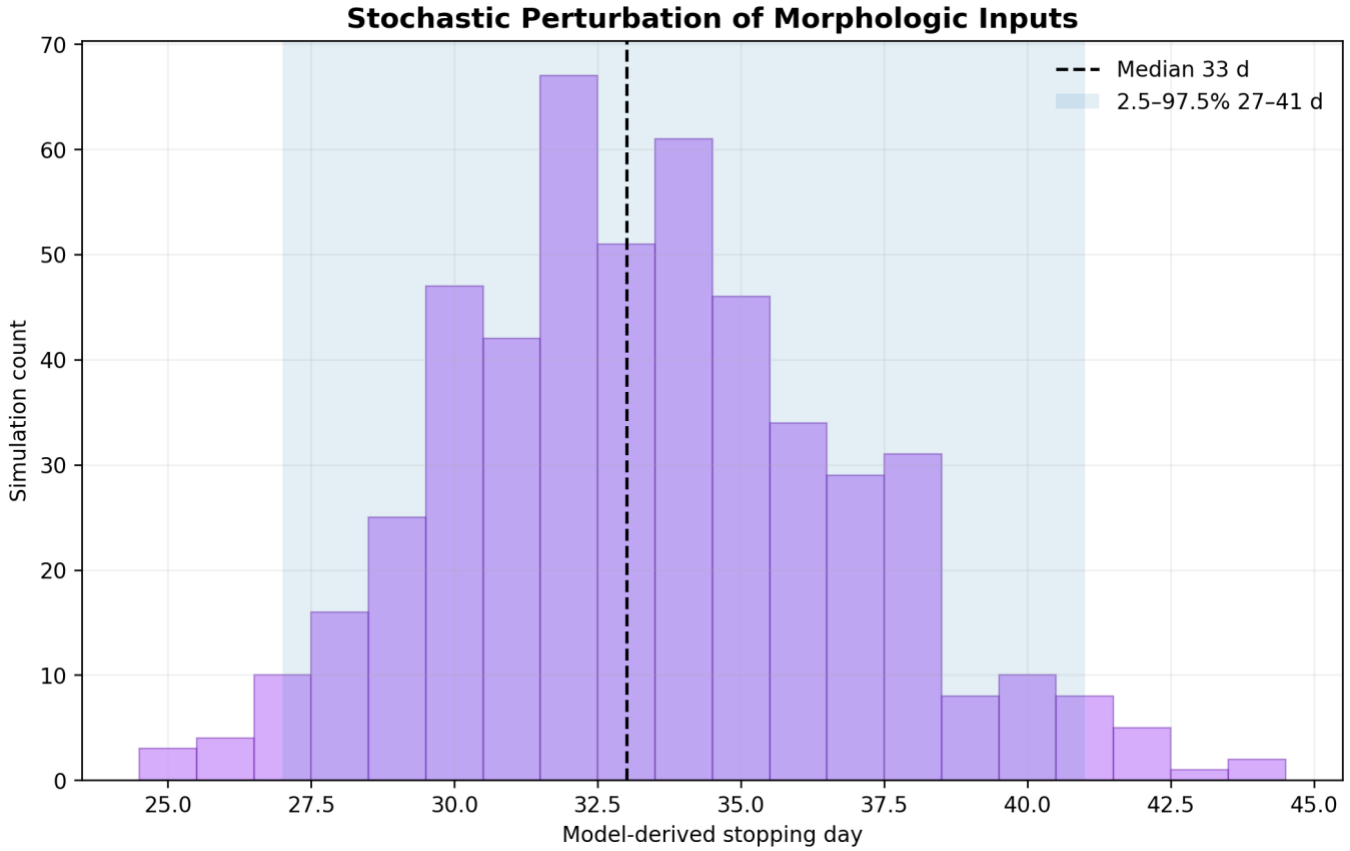


Figure 9. *Distribution of model-derived stopping days across 500 stochastic input perturbations.*

3.6. Internal Model Recovery and Numerical Robustness

In the synthetic model recovery experiment using 800 model-based states, the full dynamic state specification had the lowest AIC (156.8), followed by the morphology triad (159.9), triad plus serial enlargement (160.8), entry tear plus false lumen (397.9), and entry tear only (665.0). The corresponding internal recovery proportions were 0.958, 0.956, 0.958, 0.882, and 0.795, respectively. Because both predictors and labels were made within the model architecture, these results assess internal recoverability only and do not represent clinical predictive performance.

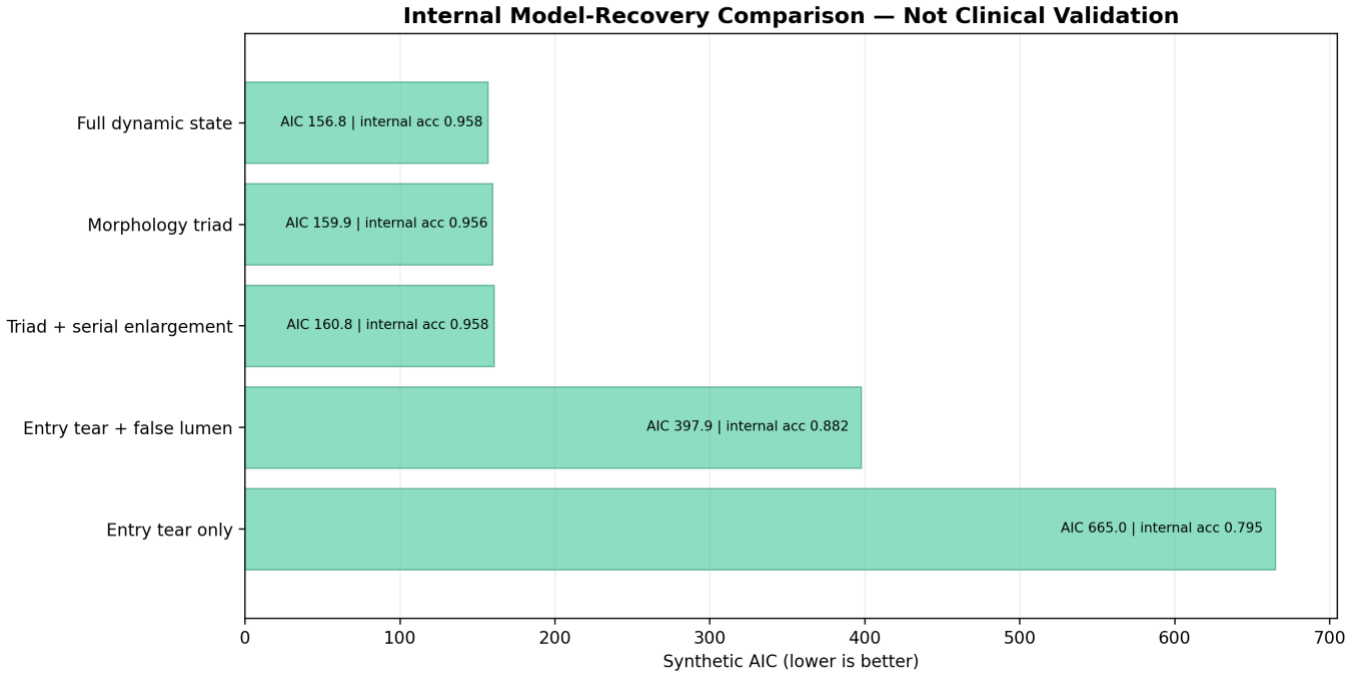


Figure 10. *AIC comparison in the internal synthetic model recovery experiment. Lower AIC indicates better relative fit to the model-based labels. This analysis is not clinical validation.*

Numerical verification showed that RK45 and DOP853 retained the 33-day reference stopping boundary across output intervals of 0.10-1.00 day, with terminal $F, D,$ and R agreeing to the displayed precision. Thus, the reported reference result was stable to the numerical solver and output intervals examined.

4. Discussion

This study interprets the behavior of a literature-informed mathematical proof-of-concept for representing surveillance versus preemptive TEVAR in uncomplicated TBAD as a state dependent optimal-stopping problem. The principal result is not the reference value of 33 days itself, but the emergence of a stopping boundary that changes with the specified morphology, latent state dynamics, and decision parameters. Across the 609-point entry tear or false lumen grid, the model-derived boundary ranged from 25 to 47 days, while the prespecified scenario analyses shifted the boundary from 26 days under combined higher risk morphology to 43 days under smaller entry tear and false lumen dimensions. These findings describe the internal behavior of the specified mathematical system and should not be interpreted as evidence that these calendar days are clinically optimal.

The directional behavior of the stopping surface is qualitatively consistent with the clinical rationale for monitoring entry tear and false lumen morphology in TBAD. Previous longitudinal studies have associated entry tear characteristics and false-lumen morphology with subsequent aortic enlargement, while contemporary guidelines incorporate morphological and serial imaging features into risk assessment (Trimarchi et al., 2014; Isselbacher et al., 2022 and Mazzolai et al., 2024). The present analysis does not independently confirm these associations, rather, it shows how literature-informed morphological variables can be incorporated into a sequential mathematical decision model. Likewise,

although previous studies have examined TEVAR and its timing (Nienaber et al., 2013; Brunkwall et al., 2014; Yang et al., 2022 and Naito and Takagi, 2026), the stopping boundaries based on here are not estimates of treatment benefit and do not establish an optimal clinical intervention window.

The uncertainty analyses further show that the stopping boundary depends materially on the assumed model parameterization. Across 500 Latin-hypercube parameter sets, the model-derived stopping day had a median of 31 days and a 2.5^{th} - 97.5^{th} percentile range of 11-50.5 days. The largest absolute PRCCs were observed for baseline waiting intensity (-0.932), acute procedural penalty (0.924), and baseline procedure intensity (0.917), indicating that the simulated boundary is particularly sensitive to provisional waiting-versus-intervention terms. In contrast, individual ±1% parameter perturbations did not alter the discretized 33-day reference boundary. The 500 stochastic input perturbations produced a median stopping day of 33 days with a 27-41-day percentile range. These quantities characterize uncertainty within the prespecified mathematical system and should not be interpreted as confidence intervals, population variability, or patient-level effect estimates.

The theoretical and computational checks support internal consistency rather than clinical validity. The specified post-TEVAR subsystem yielded five negative real Jacobian eigenvalues, supporting local asymptotic stability under the reference parameterization, while RK45 and DOP853 retained the 33-day reference boundary across the examined numerical settings. In the synthetic model recovery experiment, the full dynamic state specification had the lowest AIC among the five internally based on candidate specifications. Because the synthetic states and labels were made within the same architecture, this finding shows internal recoverability only and does not constitute predictive validation or evidence that the full specification is superior for real patients.

The contribution of model is therefore methodological rather than prescriptive. Literature-informed morphology provides interpretable reference scales, whereas $G(t), B(t), R(t)$, the Markov transition structure, theoretical waiting and procedural intensities, and Bellman costs remain model-defined components. The literature-derived sources do not numerically determine the ODE rate constants or Bellman weights. These coefficients remain provisional and were examined through sensitivity and uncertainty analyses. This modular structure allows individual assumptions to be calibrated, modified, compared, or rejected when longitudinal observations become available.

Accordingly, the immediate value of the model lies in hypothesis generation and study design rather than bedside treatment scheduling. Future calibration would require longitudinal patient level data linking serial CTA morphology, treatment timing, and outcomes, potentially including entry tear characteristics, false lumen dimensions and thrombosis, total aortic diameter, true lumen compression, serial imaging dates, blood pressure control, TEVAR timing, procedural outcomes, remodelling, reintervention, and aorta-related events. Such data would allow the observation model, state dynamics, transition structure, and decision components to be tested against observed trajectories. The present model therefore provides a reproducible and falsifiable mathematical structure for future empirical evaluation, not a validated rule for selecting the timing of TEVAR.

5. Limitations

This study has several limitations inherent to its proof-of-concept design. No patient level longitudinal outcomes were used for model fitting or validation. Therefore, the model does not establish clinical discrimination, calibration, causal treatment effects, or patient level utility. The ODE coefficients, latent state dynamics, Markov transition structure, theoretical waiting and procedural intensities, and Bellman cost parameters are provisional model components rather than effect estimates derived from the cited literature or observed patients. The model also reduces complex three-dimensional aortic anatomy to a limited set of morphological states and does not explicitly represent potentially relevant features such as tear location, re-entry tears, branch vessel involvement, true lumen compression, or false-lumen thrombosis.

Consequently, the 33-day reference boundary and the 25-47-day morphology dependent surface are simulation outputs under the specified assumptions and must not be interpreted as clinical TEVAR recommendations. Structural and practical identifiability remain unresolved because no calibrated observation model linking the latent states to longitudinal measurements has yet been established. Finally, the synthetic AIC comparison evaluates internal model recoverability only. Clinical model selection and performance measures, including AIC/BIC, discrimination, calibration, Brier score, sensitivity and specificity, decision-curve net benefit, and competing-risk performance, require appropriately designed analyses of observed patient level data.

6. Conclusion

This study developed a literature-informed mathematical proof-of-concept that formulates continued surveillance versus preemptive TEVAR in uncomplicated TBAD as a state dependent optimal stopping problem. By integrating observable aortic morphology with model-defined latent dynamics, regime switching, theoretical waiting and procedural intensities, and a finite horizon Bellman decision rule, the model based on a morphology dependent stopping boundary that varied systematically across the prespecified computational domain. Sensitivity, stochastic perturbation, equilibrium-stability, numerical-robustness, and synthetic model recovery analyses further characterized the internal behavior and uncertainty of the specified mathematical system. The resulting stopping times, including the 33-day reference boundary and the 25-47-day morphology-dependent range, are simulation outputs rather than estimates of clinically optimal TEVAR timing. The principal contribution is therefore a transparent, reproducible, and falsifiable mathematical model that can be tested, recalibrated, modified, or rejected when longitudinal patient-level CTA, treatment, and outcome data become available.

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